

## Problem 6.1

$$a) \quad cs \text{ in } S \wedge ct \text{ in } T \longrightarrow cs+ct \text{ in } S+T$$

b)  $S$  and  $T$  are two ~~skew~~ lines but  $S+T$  is a vector space, specifically a plane.

## Problem 6.2

$$\begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + y \begin{bmatrix} 3 \\ 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

## Problem 6.3

It's all linear combos of  $N(A)$  and  $N(B)$